ENERGY HACK FINAL CONCEPT

QSiteProof

Quantum-assisted + Lean-verified site-to-grid integrability in one wrapper

PSIQDK + LEAN 4

Can this site connect the requested load sooner using existing grid flexibility - and can we formally certify the resulting obligations?

QSiteProof searches site/substation/flexibility choices with PsiQDK, validates the electrical candidate with power-flow checks, then uses Lean 4 for a final GREEN/RED certificate on encoded interconnection, penalty, and permitting rules.

ONE WRAPPER - THREE ENGINES

INPUT
Site + Live Grid
Requested MW
Candidate substations
GridStatus.io signals
Flexibility + deadline

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FIND
PsiQDK
Grover / amplitude amplification over load splits, DLR, storage, curtailment, and topology.

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CHECK PHYSICS
Pandapower
Thermal limits
Voltage limits
Power flow
N-1 contingency

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PROVE
Lean 4
Capacity, flexibility, deadline, ROW/permit conditions, and penalty-trigger rules.

What PsiQDK contributes

- Searches the discrete planning space instead of quantumizing AC power flow.
- Encodes substation split, DLR, reconductoring, storage, curtailment, and topology choices.
- Returns a candidate plus FTQC resource estimates such as logical resources and Active Volume.
- Lean failures can be fed back as new constraints for the next search.

What Lean contributes

- Final trust boundary: AI/quantum search cannot self-certify.
- Proves encoded obligations only after the candidate passes physical grid checks.
- Returns the exact blocking condition when a plan is rejected.
- Produces a proof-backed GREEN/RED site-integrability result.

HACK FIT **TRACK 2 - ASSESS GRID CAPACITY AT THE EDGE | RFQ 2.2**

Does splitting one large load across several substations connect more of it?

Overall framing: X.1 - From site to plan: choose sites, check their capacity, add the flexibility needed to connect sooner (Tracks 1 + 2 + 3). QSiteProof uses PsiQDK for the RFQ 2.2 search core.

EXAMPLE FINAL CERTIFICATE

Requested capacity	PASS
Thermal / voltage	PASS
N-1 reliability	PASS
Flexibility commitment	PASS
Permit / ROW condition	PASS
Penalty trigger	FALSE

GREEN - SITE INTEGRABLE

HACK METRIC

Lean Green Yield

Compare how many candidates become formally certifiable plans per expensive verification cycle: random search vs. classical heuristic vs. PsiQDK-assisted search.

Goal: find a connect-sooner path using flexibility and existing infrastructure before defaulting to a new transmission corridor.